

Summary

Applied AI Research Engineer building and evaluating deep learning systems across computer vision, multimodal learning, and model adaptation, with particular depth in model efficiency. First-author arXiv researcher and merged Hugging Face PEFT contributor. Experienced in implementing models and training pipelines, designing controlled multi-seed experiments, debugging optimization and memory failures, and carrying research from hypothesis through evaluation and deployment benchmarking.

Core Stack PyTorch, Python, CUDA, Linux, Git

Research

- 2026 **Student Capacity Moderates Knowledge Distillation Effectiveness**, *First Author* — *arXiv:2605.31191*
Systematic Logit-KD versus Feature-KD study across four ResNet teacher–student pairs with leakage-free model selection and five-seed final experiments. Found that statistically significant KD gains occurred for higher-capacity students, while increasing teacher size at matched accuracy did not increase gains. Feature-KD matched or exceeded Logit-KD across all pairs. A controlled v2 re-run rejected an earlier bug attribution and traced the discrepancy to test-set model selection.
arxiv.org/abs/2605.31191 github.com/umutonuryasar/kd-capacity-gap

Selected Projects

- 2026 **MicroCLIP — From-Scratch Vision–Language Model**, *Independent Research Project*
Built a CLIP-style dual encoder from scratch, including a ResNet-18 image encoder, transformer text encoder, 16K BPE tokenizer, shared embedding space, contrastive objectives, deterministic training, retrieval evaluation, and zero-shot classification.
Tested SigLIP-style sigmoid loss against softmax InfoNCE in a controlled study across two objectives, batch sizes 128, 256, and 512, and three seeds on COCO Captions. Softmax matched or outperformed sigmoid at every batch size; contrary to the initial hypothesis, the gap widened as batch size decreased and sigmoid showed greater small-batch variance.
PyTorch, Vision–Language Models, Contrastive Learning, COCO, Transformers
github.com/umutonuryasar/microclip
- 2026 **RT-DETR Knowledge Distillation — COCO**, *Independent Research Project* — *v1.0 complete*
Built an end-to-end experimental framework for distilling a lightweight RT-DETR, covering five KD methods and 11 controlled configurations. Implemented transformer-specific distillation variants, loss-scale calibration, leakage-free model selection, multi-seed evaluation, regression tests, and automated result aggregation.
Best configuration improved mAP from 0.0388 to 0.0676 (+74%) in the study regime. Follow-up controls showed that loss weighting, rather than the proposed curriculum direction, explained the apparent schedule improvement.
Final fp16 student benchmark on RTX 3050: 160.8 FPS, 6.22 ms latency, and 64 MB peak VRAM with no measurable mAP loss relative to fp32.
PyTorch, Object Detection, Transformers, COCO, CUDA, FastAPI
github.com/umutonuryasar/rt-detr-kd

Open-Source Engineering

- 2026 **Hugging Face PEFT**, *Merged PR 3293*
Fixed CUDA allocator overhead in `prepare_model_for_kbit_training`, where temporary fp16-to-fp32 casts left approximately 1 GB of reserved-but-unused CUDA memory on constrained devices. Revised through maintainer review and merged into main.
github.com/huggingface/peft/pull/3293
- 2026 **aisuite**, *Merged PR 319*
Restored Python 3.14 compatibility by updating the `docstring-parser` dependency constraint away from versions relying on removed AST APIs; validated against the project test suite.
github.com/andrewyng/aisuite/pull/319

Research huggingface.co/umutonuryasar
Artifacts

Experience

- 2024–2025 **AI Trainer — Freelance**, *Outlier*, Remote
Evaluated and ranked LLM outputs for RLHF-oriented data pipelines. Authored and reviewed prompts, reference responses, and technical evaluations across programming, mathematics, and engineering.
- 2015 **Computer Vision Intern**, *HAVELSAN*, Ankara, Türkiye
Developed Python and OpenCV vision components for a flight-simulation interface and investigated ROS integration.

2014 **Avionics Intern**, *TAI — Turkish Aerospace*, Ankara, Türkiye
Developed a MATLAB computer-vision prototype for helicopter landing-zone detection.

Education

2020 **B.Sc. Electrical & Electronics Engineering**, *Bolu Abant İzzet Baysal University*, Bolu, Türkiye

Technical Skills

Deep Learning Computer vision, vision–language models, transformers, contrastive learning, representation learning, knowledge distillation

Model Efficiency PEFT and LoRA, quantization, pruning, mixed precision, memory optimization, inference benchmarking

Frameworks PyTorch, Hugging Face Transformers and PEFT, OpenCV, NumPy, scikit-learn

Systems CUDA, ONNX, TensorRT export, FastAPI, Docker, Linux

Programming Python, C/C++, Bash, MATLAB, Git, CI and testing

Languages Turkish — Native; English — Full Professional Proficiency